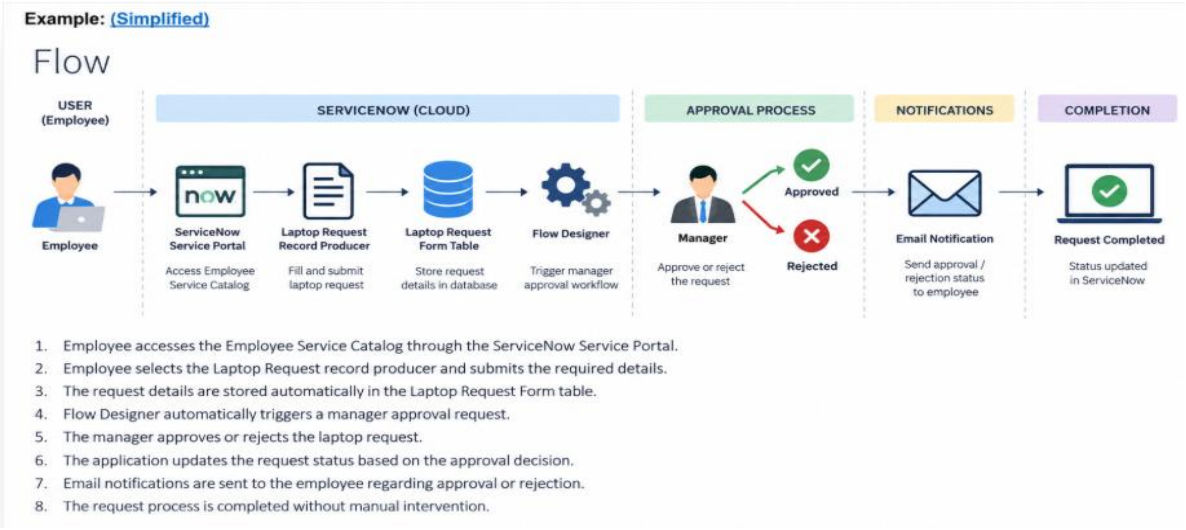
Project Design Phase-II

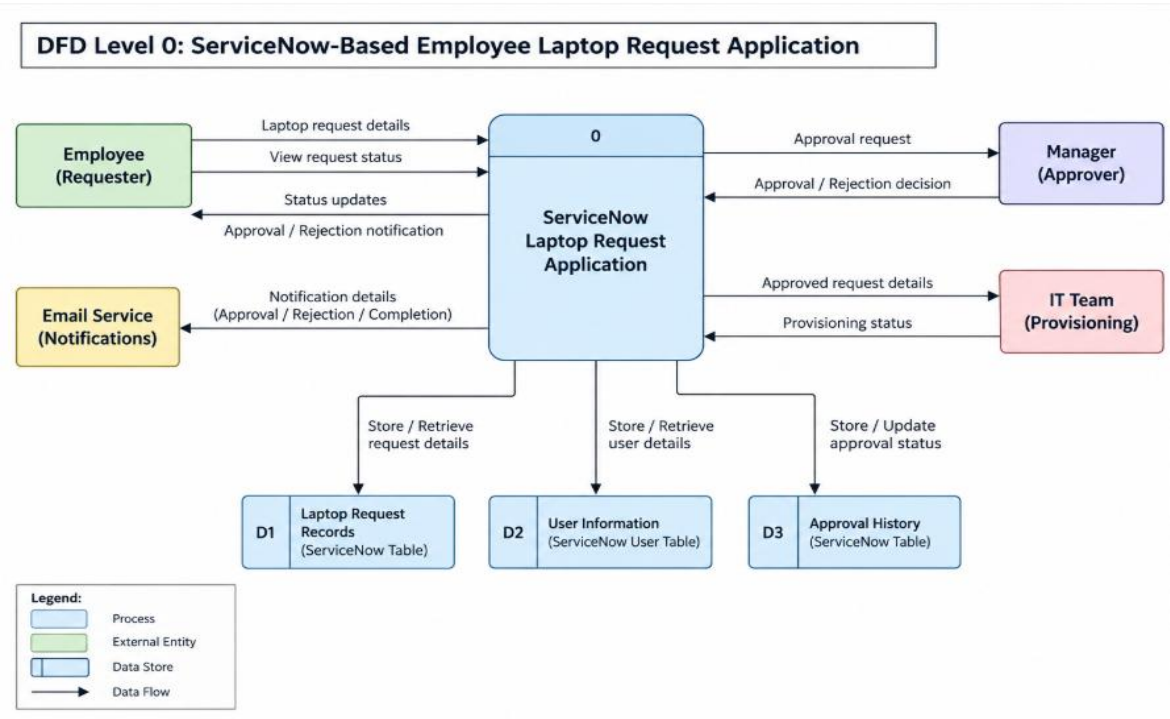
Data Flow Diagram & User Stories

Date	07 September 2026
Team ID	SWUID2026190373
Project Name	ServiceNow-Based Employee Laptop Request Application
Maximum Marks	4 Marks

Data Flow Diagram (DFD)

The Data Flow Diagram represents how information moves through the ServiceNow-Based Employee Laptop Request Application, from employee request submission through manager approval, status updates, notifications, and fulfillment.





User Stories

The following user stories describe the main functional requirements of the application.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Employee	Laptop Request Submission	USN-1	As an employee, I can access the Employee Service Catalog and submit a laptop request.	The Laptop Request item is available and the request can be submitted successfully.	High	Sprint-1
Employee	Request Details	USN-2	As an employee, I can provide requester, email, required date, laptop type, reason, manager and short description.	All required variables are captured and mapped to the correct Laptop Request Form fields.	High	Sprint-1
System	Record Creation	USN-3	As the system, I automatically create a Laptop Request Form record when the Record Producer is submitted.	A new record is created in u_laptop_request_form with the submitted values.	High	Sprint-1
Manager	Manager Approval	USN-4	As a manager, I can receive and approve or reject an employee laptop request.	The manager is automatically assigned as approver and can make an approval decision.	High	Sprint-1
System	Approval Automation	USN-5	As the system, I automatically trigger the approval process when a new laptop request record is created.	Flow Designer starts on record creation without manual intervention.	High	Sprint-1
Employee	Approval Notification	USN-6	As an employee, I receive a notification when my laptop request is approved.	Status is updated to Approved and an approval email is sent to the requester.	High	Sprint-1
Employee	Rejection Notification	USN-7	As an employee, I receive a notification when my laptop request is rejected.	Status is updated to Rejected and a rejection email is sent to the requester.	High	Sprint-1
IT / Administrator	Request Tracking	USN-8	As an administrator or fulfillment user, I can view and track laptop request records and their current status.	Request records contain the relevant details, manager decision, and current status.	Medium	Sprint-2

Automation Scope: The application automates the end-to-end flow: record creation → manager approval → approval decision handling → status update → email notification → fulfillment tracking.

Project Design Phase-II
Technology Stack (Architecture & Stack)

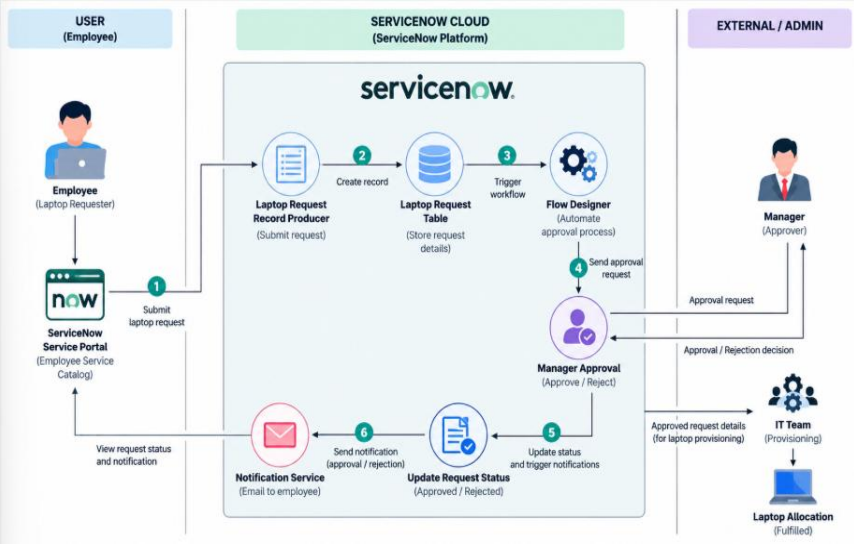
Date	07 September 2026
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Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Technical Architecture: Employee → ServiceNow Service Catalog → Record Producer → Laptop Request Form Table → Flow Designer → Manager Approval → Status Update → Email Notification → Request Tracking

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



Guidelines:

Show the main **ServiceNow processes and approval flow**.
Separate **Local users and ServiceNow Cloud**.
Include **Email notifications and data storage**.
Machine Learning: Not Applicable.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Employee-facing catalog and request form interface.	ServiceNow Service Portal / Employee Center
2	Catalog & Record Producer	Creates laptop request records from catalog variables.	ServiceNow Service Catalog and Record Producer
3	Application Logic-1	Stores and manages laptop request information.	Custom Table: u_laptop_request_form
4	Application Logic-2	Automates approval and request lifecycle.	ServiceNow Flow Designer
5	Approval Management	Routes the request to the selected manager and records the decision.	ServiceNow Approvals / sysapproval_approver
6	Email Notification	Sends request and decision notifications.	ServiceNow Notifications / Flow Designer
7	Database	Stores request fields, statuses, and approval data.	ServiceNow Platform Database
8	Security & Access	Controls who can create, approve, view, and manage requests.	ServiceNow Roles and ACLs
9	External Interface	Uses ServiceNow user records and email services for routing and communication.	sys_user and Email Services
10	Infrastructure (Cloud)	Hosts the application and automation.	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Platform Framework	Built using native ServiceNow application and automation capabilities.	ServiceNow Service Catalog, Record Producer, Flow Designer
2	Security Implementations	Access is controlled through authentication, roles, ACLs, and approval permissions.	ServiceNow Authentication, Roles, ACLs

S.No	Characteristics	Description	Technology
3	Scalable Architecture	Reusable catalog, table, and workflow components support growth in users and requests.	ServiceNow Cloud Platform
4	Availability	The application uses managed cloud infrastructure provided by the ServiceNow platform.	ServiceNow Cloud Infrastructure
5	Performance	Automation reduces manual processing and enables direct routing, updates, and notifications.	Flow Designer and Platform Services

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d

Editable Technical Architecture Flow

Employee	Service Catalog	Record Producer	Laptop Request Table	Flow Designer
Manager Approval	Approval Decision	Status Update	Email Notification	Request Tracking

EMPLOYEE EXPERIENCE

Employee Laptop Request

Customer Journey Map

A seamless and transparent process for employees to request, track and receive their work laptop through ServiceNow.



SCENARIO

Requesting, approving, procuring and receiving a laptop for official use

	Entice	Enter	Engage	Exit	Extend
	How does the employee become aware of the process?	What do they do to start the process?	What happens during the request and approval process?	What do they experience when the request is completed?	What happens after the laptop is received?
Steps What does the person or group typically experience?	Learn about the laptop request process Access ServiceNow Service Portal	Open Laptop Request form Fill in required details Submit the request	Manager reviews request Approval / Rejection decision Request sent to IT for provisioning	Receive notification Track request status Receive laptop	Provide feedback (if required) Request support or replacement in future Process improvements
Interactions What interactions do they have at each step? • **People:** Who do they talk to? • **Places:** Where are they? • **Things:** What systems or objects do they use?	Email / company communication about the process SaptooNour Service Portal (Employee Service Catalog)	Laptop Request Record Producer Employee fills in details (laptop type, purpose, location etc.) Submit button in ServiceNow	Manager receives approval task in ServiceNow Manager approves or rejects the request Automatic workflow to IT Team (Flow Designer)	Email notification to employee Check status in ServiceNow (My Requests) Collect laptop from IT / delivery to location	Rate the experience / provide feedback Raise a support ticket if there are issues Receive communication for future upgrades or replacements
Goals & motivations At each step, what is a person's primary goal or motivation? ("Help me..." or "Help me avoid...")	Help me know how to request a laptop Help me access the right portal easily	Help me submit my laptop request quickly and accurately Help me provide all required information Help me track my request	Help me get my request approved Help me know the status of my request Help me ensure timely provisioning	Help me receive confirmation of approval Help me know when I will receive my laptop Help me get the laptop without delays	Help me share my feedback Help me get support if there are issues Help me have a better and faster experience in future
Positive moments What steps are enjoyable, productive or satisfying?	Clear information about the process Easy access to the portal	Simple and user-friendly form Quick submission and confirmation message	Timely approval from manager Real-time status updates Smooth handover to IT team	Notification received promptly Laptop received on time Ready to use the laptop for work	Option to give feedback Responsive support for issues Continuous improvements based on feedback
Negative moments What steps are frustrating, confusing or time-consuming?	Not aware of the process Difficulty finding the right portal	Form is complex or unclear Missing required information Submission errors	Delay in manager approval Lack of status updates Request stuck in process	Delay in receiving notification Laptop delivery takes longer than expected Issues with laptop configuration	No follow-up after request completion Difficulty in raising support requests Lack of visibility on future upgrade options
Areas of opportunity How might we make each step better? What ideas do we have?	Better awareness through onboarding and emails Direct link to the ServiceNow portal	Simplify the form with clear instructions Mandatory field validations Confirmation page with request number	Automated reminders for managers Real-time status tracking and notifications SLA-based provisioning process	Instant notifications at each stage Estimated delivery date in portal Self-service tracking and updates	In-app feedback option Knowledge base for common issues Periodic communication on upgrade cycles

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	07 September 2026
Team ID	SWUID2026190373
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Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Laptop Request Submission	Employee submits a laptop request through the Employee Service Catalog using the Laptop Request record producer.
FR-2	Request Data Capture	Captures requester, employee email, required date, short description, status, reason, type of request, and manager.
FR-3	Automatic Manager Approval	When a request record is created, Flow Designer automatically sends an approval request to the selected manager.
FR-4	Approval Status Handling	Updates the request status to Approved or Rejected based on the manager's approval decision.
FR-5	Email Notifications	Sends notifications to the requester when the request is submitted, approved, or rejected.
FR-6	Request Tracking	Allows authorized users to view the request details and current request status.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The catalog form and approval process should be simple, clear, and easy to use.
NFR-2	Security	User roles and access controls should protect request and approval information.
NFR-3	Reliability	Request creation, approval, and status updates should execute consistently without data loss.
NFR-4	Performance	The form, approval flow, and notifications should complete within acceptable response times.
NFR-5	Availability	The application should be accessible whenever the ServiceNow platform is available.
NFR-6	Scalability	The solution should support increasing employees and laptop requests using reusable platform components.